

**NATIONAL COLLEGE OF SCIENCE AND TECHNOLOGY**

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**Computer Studies Department**

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**“AutoRefill: Design and Development of Water Vendo Machine Refiller System”**

**A Capstone Project**

Presented to the Faculty of

Computer Studies Department

**National College of Science and Technology**

**Dasmariñas, Cavite**

In Partial Fulfillment

of the Requirements for the Degree

**Bachelor of Science in Information Technology**

by

**DENZ DECLARADO**

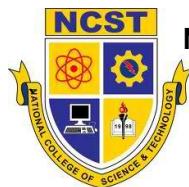
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**I. INTRODUCTION**



Access to clean and affordable drinking water is essential in many communities.

Water vendo machines are commonly used in residential areas because they provide convenient and low-cost water refilling services. However, many vendo machines still rely on manual monitoring of water levels, which can lead to delays in refilling, service interruptions, and inefficient operations.

Target Client / Beneficiary (e.g., barangay, school, SME, department)

Local water refilling station owners, Small and Medium Enterprises (SMEs) operating water vendo machines, Barangay communities using public water vending stations

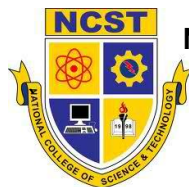
General Overview of the Problem

Most water vendo machines require manual checking and refilling of water storage tanks. When operators fail to monitor levels regularly, machines may run out of water, causing inconvenience to users and loss of revenue for operators. There is a need for an automated system that can monitor and refill water levels efficiently.

## **II. PROJECT TITLE**

Exact Title of the Capstone/Thesis Project

AutoRefill: Design and Development of Water Vendo Machine Refiller System



Must be clear, concise, and reflective of the system or research focus

The title clearly reflects the development of an automated dispensing system focusing on accuracy, efficiency, and microcontroller integration.

### **III. PROJECT CONTEXT / BACKGROUND OF THE STUDY**

What is the situation that led you to propose the project?

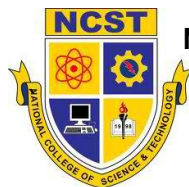
To recurring issue of water vendo machines running out of water due to delayed or inconsistent monitoring by operators. In many cases, personnel must physically inspect the tanks to determine whether refilling is needed. This manual process is time-consuming, labor-intensive, and prone to human error, particularly in high-demand locations where water consumption is frequent. As a result, customers may experience inconvenience when machines become temporarily unavailable.

Describe the current process/system of the client:

1. The operator places the water container manually under the dispensing outlet.
2. The pump or valve is manually activated to start the water flow.
3. The operator estimates the appropriate filling level and manually stops the process.

Include brief industry background if applicable

Automated monitoring and refilling systems can now be developed to optimize operations, reduce downtime, and improve service reliability. This project aims



to create a prototype system that automatically detects water levels and activates the refill mechanism when needed.

#### **IV. STATEMENT OF THE PROBLEM**

##### **General Problem**

How can an automated system be developed to efficiently monitor and refill water levels in a water vendo machine to ensure continuous operation?

Specific Problems (Enumerate the issues that your project aims to solve)

1. Manual monitoring of water levels causes delays and inefficiency.
2. Water vendo machines may run out of water unexpectedly.
3. Operators lack a real-time indication of tank status.
4. Service interruptions lead to customer dissatisfaction and revenue loss.

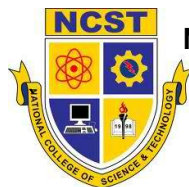
#### **V. OBJECTIVES OF THE STUDY**

##### **General Objective**

To design and develop an Arduino-based automated refiller system for water vendo machines that ensures continuous and efficient operation.

Specific Objectives (Should align with each specific problem)

1. To create a water level monitoring system using sensors.
2. To develop an automatic refill mechanism controlled by Arduino.



3. To provide visual or alert indicators for tank status.
4. To test the functionality and reliability of the prototype.

## **VI. SCOPE AND LIMITATIONS**

What features or modules are included?

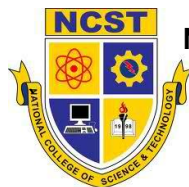
The project will focus on designing a working prototype. It will include water level detection, automatic pump control, and status indicators. The system will be tested in a simulated or small-scale environment.

The system will include the following features:

1. Automated water dispensing mechanism
2. Precise volume measurement using sensors (e.g., flow sensor or load sensor)
3. User input for selecting desired water volume
4. Pump and valve control module
5. Real-time status display (LCD/OLED or indicator lights)
6. Prototype hardware implementation and testing

What are the boundaries of the system?

1. Monitoring the water level in a single tank and automatically refilling it when needed.
2. Dispensing water in precise volumes selected by the user.
3. Display real-time status through an LCD/OLED screen or indicator of lights.
4. Controlling the water flow using an Arduino-controlled pump and solenoid valve.



5. Testing and validation conducted in a simulated or controlled environment.

What are the constraints, and what's excluded?

Constraints

1. The system relies on low-cost microcontrollers and sensors, so accuracy may be limited compared to industrial-grade solutions.
2. The prototype is designed for small volumes; large commercial-scale tanks may require modification.
3. Continuous operation for extended periods is not tested; the system is primarily for proof-of-concept purposes.

Exclusions

1. Water quality monitoring (e.g., pH, TDS, or contamination detection)
2. Water purification or filtration is not part of the system.
3. Multi-tank or multi-machine management is outside the scope.
4. Full-scale commercial deployment, advanced security features, or automated payment systems are not included.

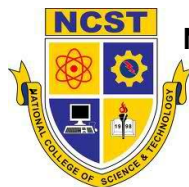
## **VII. SIGNIFICANCE OF THE STUDY**

Who will benefit from this project?

Vendo machine operators, Customers, Communities, Future researchers

How will it help the client or users?

1. Reduces Manual Effort. Operators no longer need to constantly check and refill the water tank manually, saving time and labor.



2. Ensures Continuous Service. Automatic refilling prevents the machine from running out of water, ensuring customers always have access.
3. Improves Accuracy. Sensors and automated control allow precise water dispensing, minimizing overfilling or spillage.
4. Enhance Monitoring. Real-time status indicators inform the operator of water levels and dispensing progress.
5. Supports small businesses. By improving efficiency and reliability, operators can maintain better customer satisfaction and potentially increase revenue.

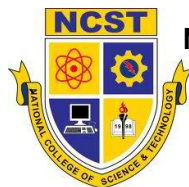
## **VII. TARGET OUTPUT / EXPECTED FEATURES**

What will be delivered?

- Functional automated water vendo machine refill prototype
- System documentation including design, features, and testing results

Initial list of system modules or functionalities

- Water level sensing module
- Automatic pump control module
- Indicator display (LED/LCD)
- Power supply and safety controls



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Tools/technologies to be used (e.g., PHP, MySQL, Python, etc.)

1. Arduino microcontroller
2. Water level sensors / float sensors
3. Relay module
4. Water pump
5. LCD or LED indicators
6. Embedded C / Arduino IDE